



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 14



BARANGAY Corn Mill
Model: CM-250

BARANGAY CM-250 CORN MILL

Test Report No.	:	2013 – 14
Date of Test	:	February 7, 2013
Test Valid Until	:	February 7, 2018
Brand	:	BARANGAY
Model	:	CM-250
Type	:	Steel huller type milling units
Designed and Fabricated by	:	BALBINO C. GERONIMO Bay, Laguna
Test Requested by	:	BALBINO C. GERONIMO
Machine Classification	:	Commercial

SUMMARY

The BARANGAY CM-250 Corn Mill had the following capacities: Input Capacity of 187.5 kg/h corn kernel; Output Capacity of 128.8 kg/h corn grits; and Milling Capacity of 180.0 kg/h corn kernels using IPB VAR 6 variety with Moisture Content of 12.5 %, wet basis. Test results showed that the corn mill had a Main Product Recovery of 68.7 % and By-Product Recovery of 23.6% with Losses of 7.7%.

Laboratory analysis of samples taken from the output chutes of the sifter assembly showed the following results: 1) output from first screen had 80.4% grit #10; 16.8% grit #12 and grit #14; 2.6% grit #16 and grit#18; 0.1% grit #20 and grit #24; and, 0.1% floured corn, germ and bran; and, 2) output from second screen had none grit #10; none grit #12 and grit #14; 11.9% grit #16 and grit#18; 46.8% grit #20 and grit #24; and, 41.3% floured corn, germ and bran.

The average Fuel Consumption of the SUPER KAMA 178F diesel engine which drove all the rotating components of the corn mill was 0.86 L/h.

During the tests, two persons were needed to operate the machine.

PURPOSE AND SCOPE OF TEST

The machine was tested to determine its performance. The test was conducted at the Institute of Plant Breeding (IPB), University of the Philippines Los Baños, College Laguna on February 7, 2013 and was evaluated on the following items:

1. Capacities
 - a) Input Capacity
 - b) Output Capacity
 - c) Milling Capacity
2. Milling Recoveries
 - a) Main Product Recovery
 - b) By-Product Recovery
3. Losses
3. Fuel Consumption
4. Noise Level
5. Labor Requirement
6. Quality of Work

METHOD OF TEST

The test was based on the Philippine Agricultural Engineering Standard “PAES 211:2000 Agricultural Machinery – Corn Mill – Methods of Test”.

OPERATOR'S MANUAL

No operator's manual or brochure was submitted.

DESCRIPTION OF THE MACHINE

The BARANGAY CM-250 Corn Mill as shown in **Figure 1** was a steel huller-type corn mill which consisted of the following components: hopper, milling units, oscillating sieve assembly, fan, and prime mover. The machine was powered by a 5.22 kW SUPER KAMA KM178F diesel engine (**Figure 2**) which drove its belt-pulley transmission system (**Figure 3**).

The hopper (**Figure 4**) which was located on top of the milling unit held the corn kernel for milling. The rate of feeding was controlled by a sliding gate located at the bottom of the hopper (**Figure 5**).

From the hopper, the grains pass through two milling units (**Figure 6**). The steel huller was a solid steel cylinder (**Figure 7**) with a screwed portion which served as feeder to the milling chamber. The other portion of the solid steel cylinder was provided with rectangular-shaped protrusions. An adjustable blade and a screen below the steel cylinder were installed inside the milling chamber. Corn kernels were reduced into corn grits and the germs and corn pericarp were removed inside the milling chamber. The retention time of the corn kernel inside the milling chamber was controlled by the adjustable sliding counter weight at the grain outlet spout where ground corn/corn grits were discharged.

From the second milling unit, the grains pass through an oscillating sieve assembly (**Figure 8**) consisting of upper and lower screens with different screen mesh for separation of corn grits to different sizes and the bran. The oscillating sieve assembly was provided with three outlet chutes (**Figure 9**): first, for the grits that did not pass the first sieve and recycled back to the milling units, second, for the upper sieve; and, third, for the lower sieve.

The corn mill was also provided with a centrifugal type suction fan (**Figure 10**) which was used to suck the germ, corn pericarp and impurities from the outlet chute of the milling units and the output chute.

The main frame of the machine was made of angle bar materials.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. BARANGAY Corn Mill
Model: CM-250

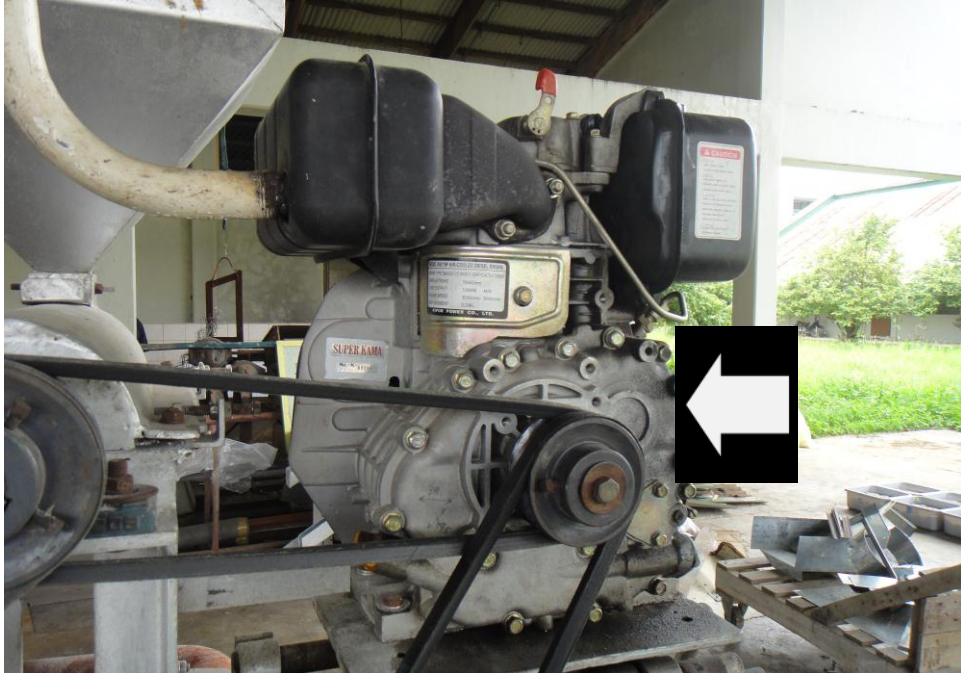


Figure 2. Corn mill's prime mover

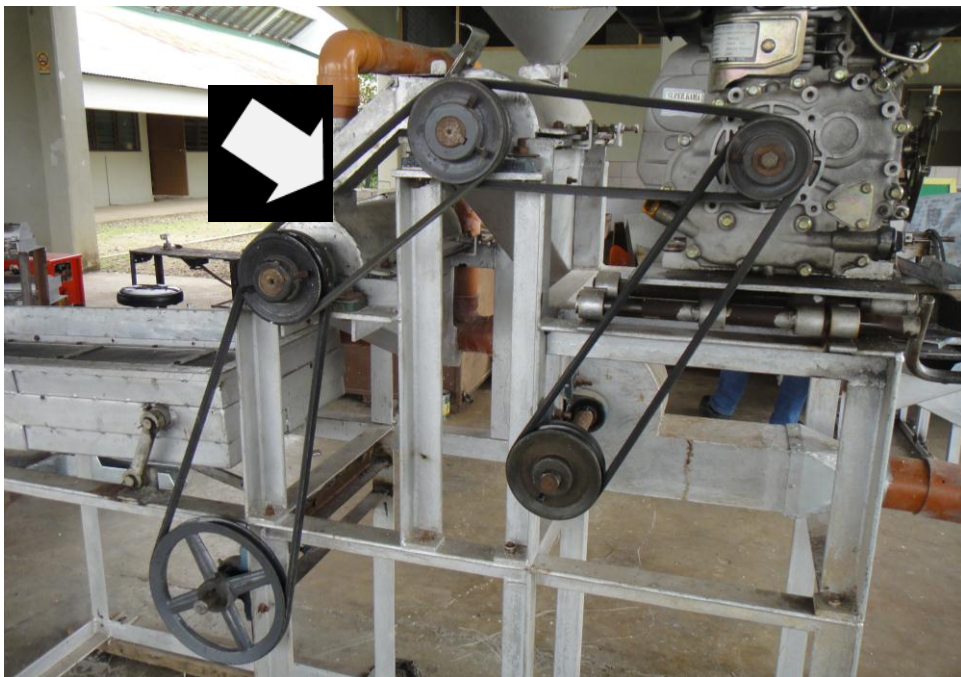


Figure 3. Belt-pulley transmission system



Figure 4. Hopper

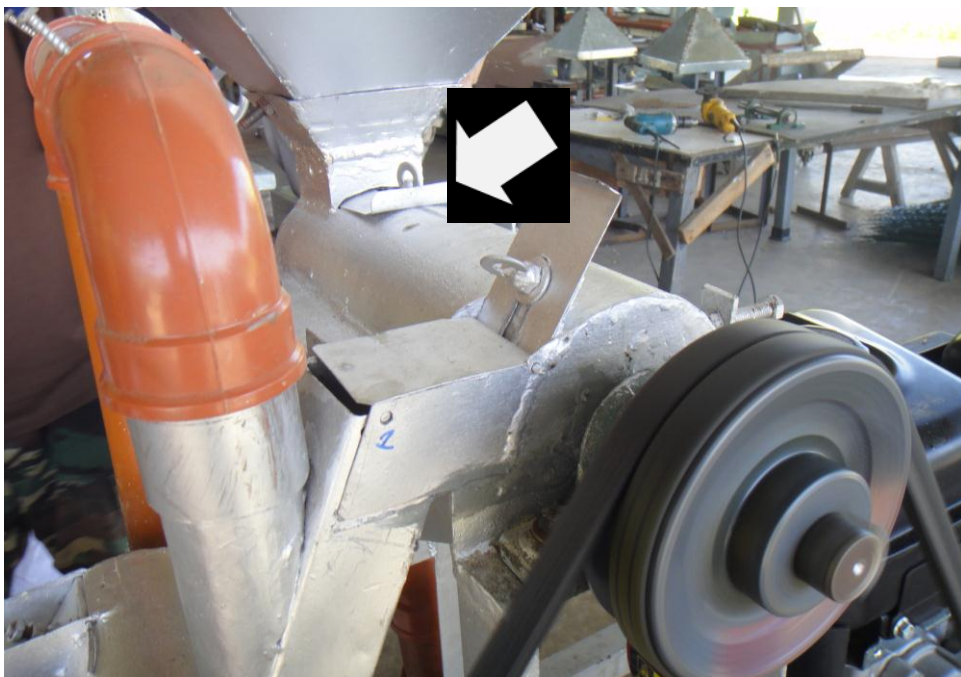


Figure 5. Sliding Gate

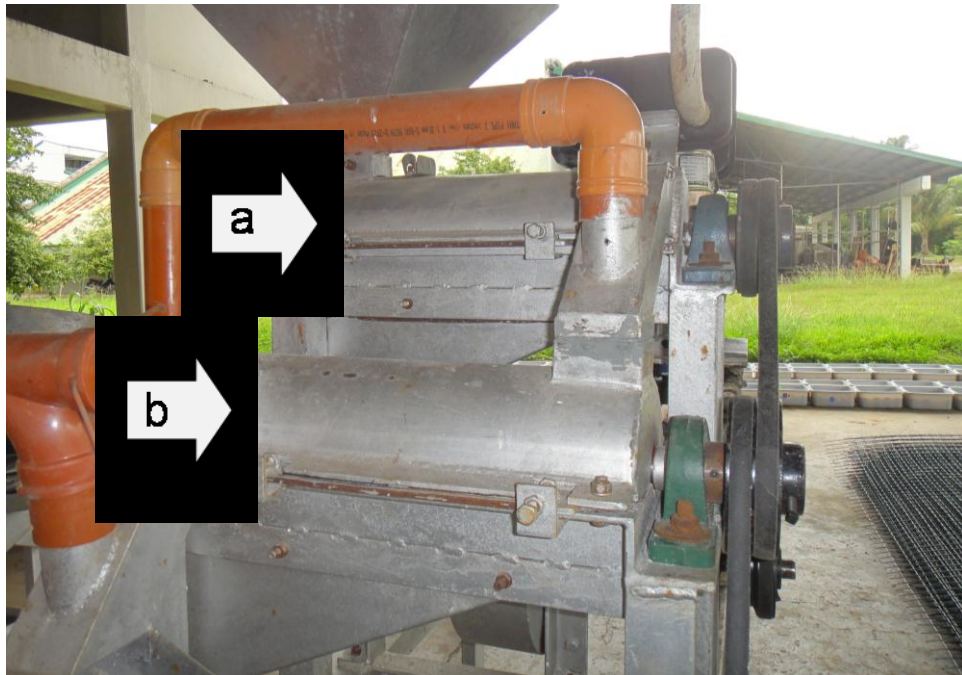


Figure 6. a) First Milling Unit
b) Second Milling Unit

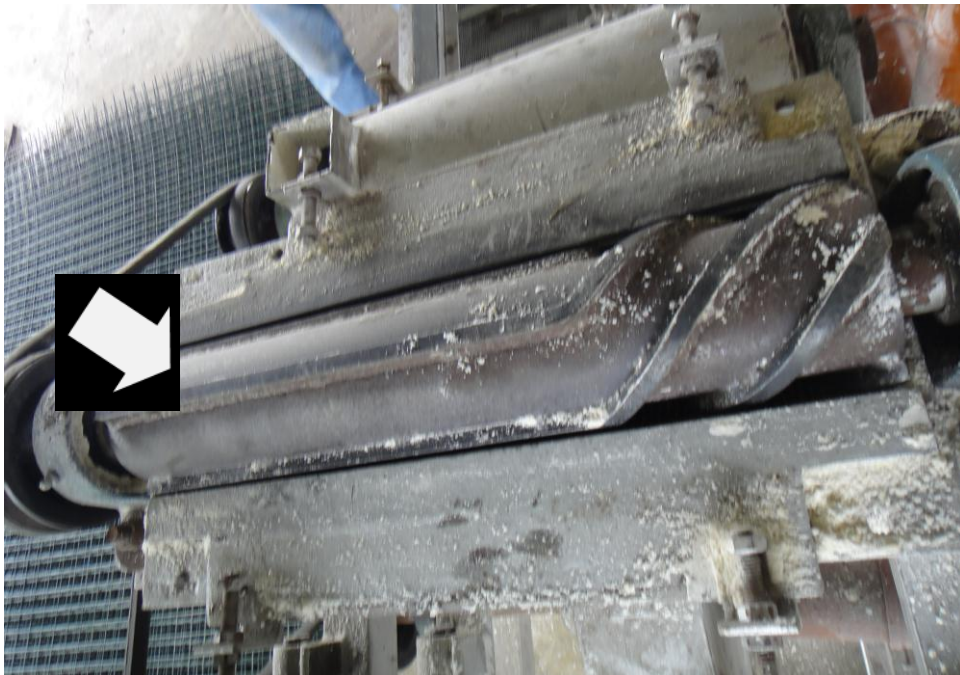


Figure 7. Solid Steel Cylinder

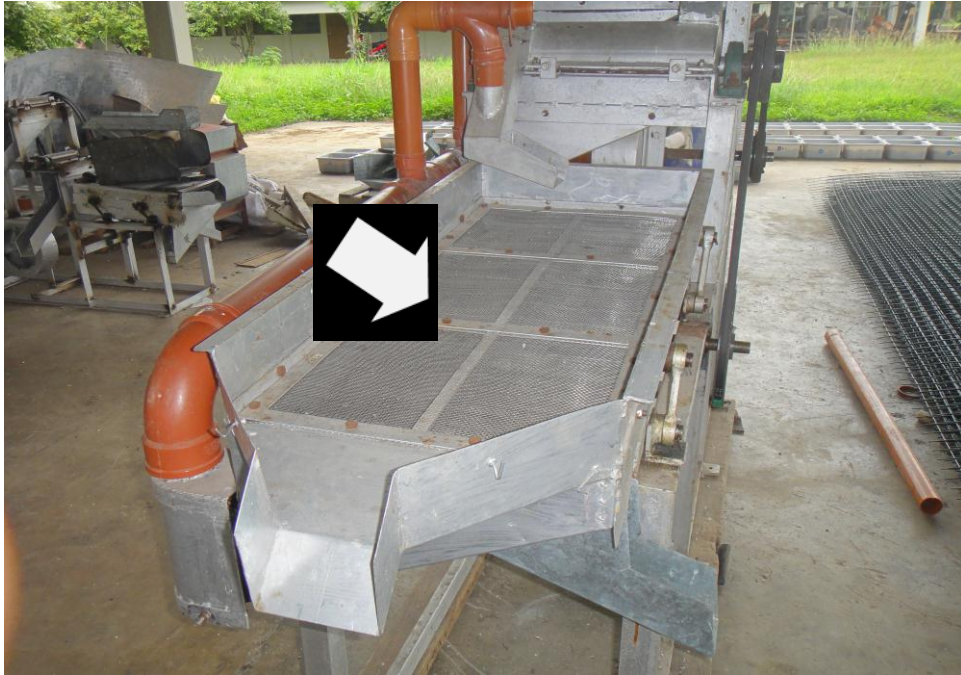


Figure 8. Oscillating Sieve Assembly

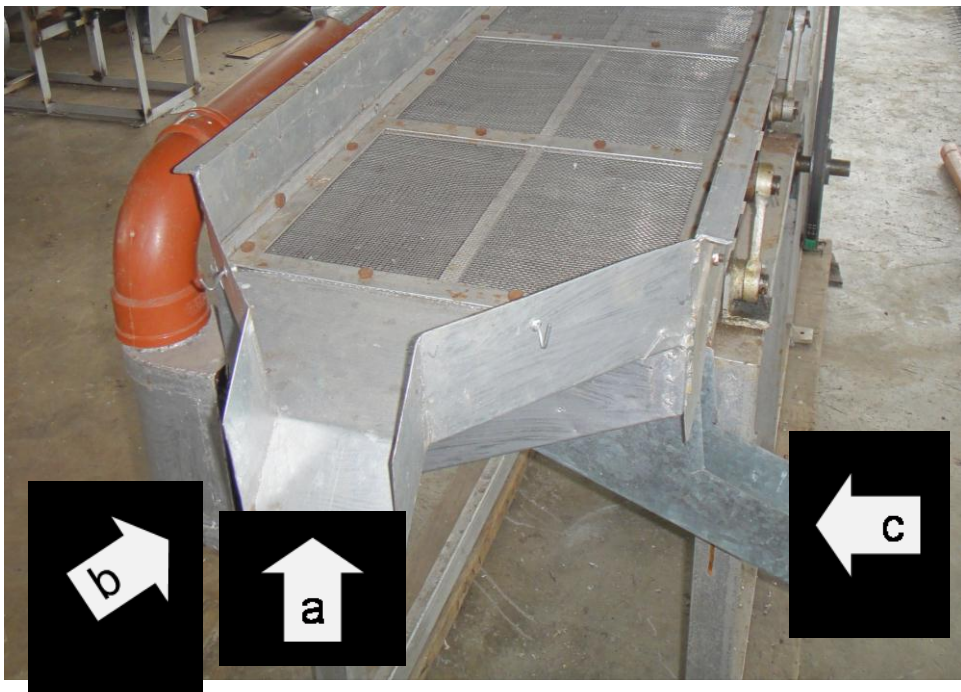


Figure 9. a) Outlet chute for grits that did not pass the upper screen
b) Upper screen outlet chute
c) Lower screen outlet chute

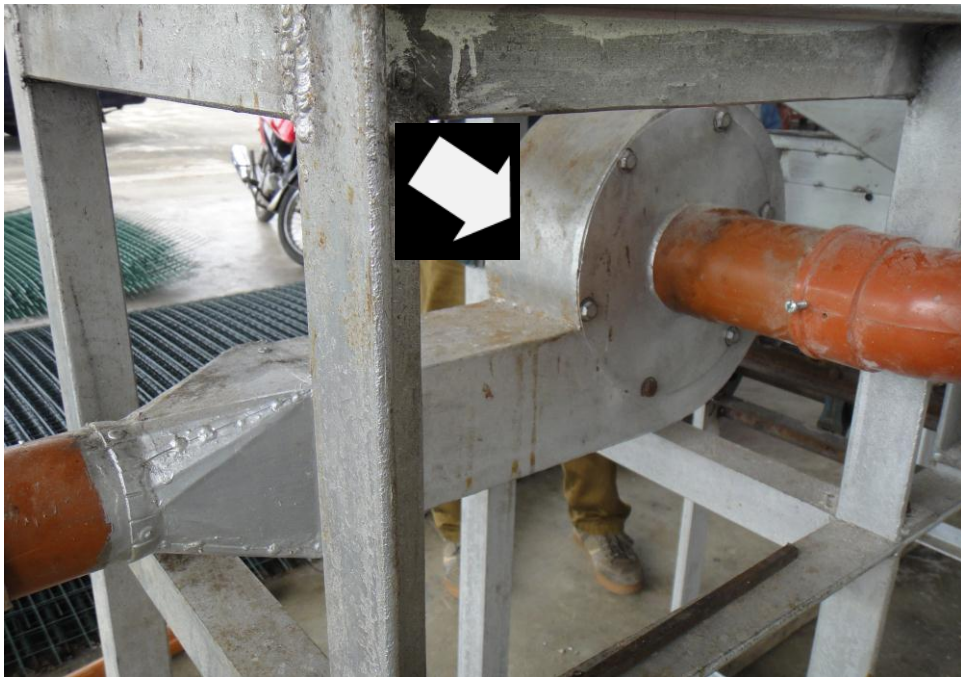


Figure 10. Suction fan

MACHINE SPECIFICATIONS

Table 1. Detailed specifications of the BARANGAY CM-250 Corn Mill

A.	Brand	:	BARANGAY
B.	Model	:	CM-250
C.	Serial No.	:	No Data
D.	Designer and Fabricator	:	Balbino C. Geronimo Bay, Laguna
E.	Overall Dimensions and Weight		
1.	Length (mm)	:	2140
2.	Width (mm)	:	735
3.	Height (mm)	:	1725
4.	Weight (kg) without prime mover	:	No Data
F.	Hopper		
1.	Bottom opening (L x W), mm	:	50 x 50
2.	Height from the ground, mm	:	1725
3.	Material	:	G. I. Sheet, 1.0 mm thick
G.	First (upper) Steel Huller Mill Assembly		
1.	Housing material	:	G.I. Pipe, Sch. 20
2.	Solid steel cylinder		
a)	Size (L x D), mm	:	435 x 93
b)	Length of portion with horizontal protrusion, mm	:	305
c)	Number of horizontal protrusions	:	4
H.	Second (lower) Steel Huller Mill Assembly		
1.	Housing material	:	G.I. Pipe, Sch. 20

Table 1, Detailed specifications of the BARANGAY CM-250 Corn Mill (Cont'n.)

2.	Solid steel cylinder		
a)	Size (L x D), mm	:	435 x 93
b)	Length of portion with horizontal protrusion, mm	:	305
c)	Number of horizontal protrusions	:	4
I.	Counter Blades		
1.	Dimensions (L x W x T), mm	:	435 x 78 x 6.5
2.	No. of pieces/hulling assembly	:	2
3.	Material	:	C1045
J.	Hexagonal screen		
1.	Configuration	:	Half-moon
2.	Size of perforations (L x W), mm	:	10 x 2
K.	Oscillating Sieve Assembly		
1.	Size (L x W), mm	:	1150 x 490
2.	No. of sieve layers	:	2
3.	Size of perforations (upper sieve), mm	:	3 x 3
4.	Size of perforations (lower Sieve), mm	:	1.5 x 1.5
5.	Material (screen)	:	Wire Mesh
L.	Aspirating Fan		
1.	Housing		
a)	Size, W x D (mm)	:	275 x 110
b)	Material	:	G. I. Sheet, 1.0 mm thick
2.	Blade		
a)	No. of pieces	:	6
b)	Size (L x W), mm	:	102 x 64
c)	Material	:	B. I. Sheet, 2 mm thick

Table 1, Detailed specifications of the BARANGAY CM-250 Corn Mill (Cont'n.)

M.	Main Frame	:	Angle Bar (51 mm x 51 mm x 6 mm)
N.	Power Transmission System		
1.	Engine to upper steel huller shaft	:	Double groove, B-section pulley and V-belt
	Engine pulley	:	89 mm x 2B x 25 mm
	Steel huller shaft pulley	:	127 mm x 2B x 30 mm
	Belt	:	B - 40
2.	Engine to blower shaft	:	Double groove, B-section pulley and V-belt
	Engine pulley	:	89 mm x 2B x 25 mm
	Blower shaft pulley	:	127 mm x 2B x 25 mm
	Belt	:	B - 48
3.	Upper steel huller shaft to lower steel huller shaft	:	Double groove, B-section pulley and V-belt
	Upper steel huller shaft pulley	:	127 mm x 2B x 38 mm
	Lower steel huller shaft pulley	:	127 mm x 2B x 38 mm
	Belt	:	B-39
4.	Lower steel huller shaft to oscillating sieve shaft	:	Single groove, B-section pulley and V-belt
	Lower steel huller shaft pulley	:	127 mm x 2B x 38 mm
	Oscillating sieve shaft pulley	:	254 mm x 1B x 25 mm
	Belt	:	B-65
O.	Engine (Prime mover)		
1.	Brand	:	SUPER KAMA
2.	Model	:	KM178F
3.	Make	:	WUXI WORLDBEST KAMA POWER CO., LTD., Shanghai, China

Table 1, Detailed specifications of the BARANGAY CM-250 Corn Mill (Cont'n.)

4.	Maximum rated power (kW)	:	5.22 @ 3600 rpm
5.	Continuous rated power (kW)	:	2.98 @ 3600 rpm
6.	Type	:	Four-stroke, air-cooled, single inclined cylinder diesel engine
7.	Displacement (cc)	:	296
8.	Starting system	:	Rope (recoil)
9.	Cooling system	:	Air-cooled
10.	Weight (kg)	:	32.0

RESULTS AND DISCUSSIONS

The BARANGAY CM-250 Corn Mill which basically consisted of two steel hullers milling units was tested using IPB Var 6 corn variety. The general information of the test material is shown in **Table 2** and the results of the performance test are shown in **Table 3**.

Input Capacity

The Input Capacity of a corn mill refers to the amount of corn kernels fed to the corn mill per unit of time. The machine had an average Input Capacity of 187.5 kg/h.

Output Capacity

The Output Capacity of a corn mill refers to the amount of corn grits output of the corn mill per unit of time. The machine had an average Output Capacity of 128.8 kg/h.

Milling Capacity

The Milling Capacity of a corn mill refers to the total amount of corn kernels milled over the total time the corn mill is in operation. The machine had an average Milling Capacity of 180.0 kg/h.

Main Product Recovery

The Main Product Recovery was the ratio of the main product (corn grits #10, #12, #14, #16 and #18) to the total weight of corn kernel input, expressed in percent. The machine had a Main Product Recovery of 68.7 %.

By-Product Recovery

The By-Product Recovery was the ratio of the by-product (corn grits #20 and #24, floured corn, germ and bran) to the total weight of corn kernel input, expressed in percent. The machine had a By-Product Recovery of 23.6 %.

Degerminator Efficiency

The Degerminator Efficiency was the ratio of the degerminated (shelled corn kernels whose germ and pericarp have been removed) corn kernel sample to the initial weight of sample. The machine had Degerminator Efficiency of 5.3%.

Fuel Consumption

The 5.22 kW SUPER KAMA KM178F diesel engine which drove all the rotating components of the corn mill had a Fuel Consumption Rate of 0.86 L/h.

Labor Requirement

During the tests, two persons (**Figure 11**) were needed to operate the corn mill.

Observation

- 1) The corn mill belt-pulley transmission system was not provided with belt cover (**Figure 12**).



Figure 11. Operators of the corn mill

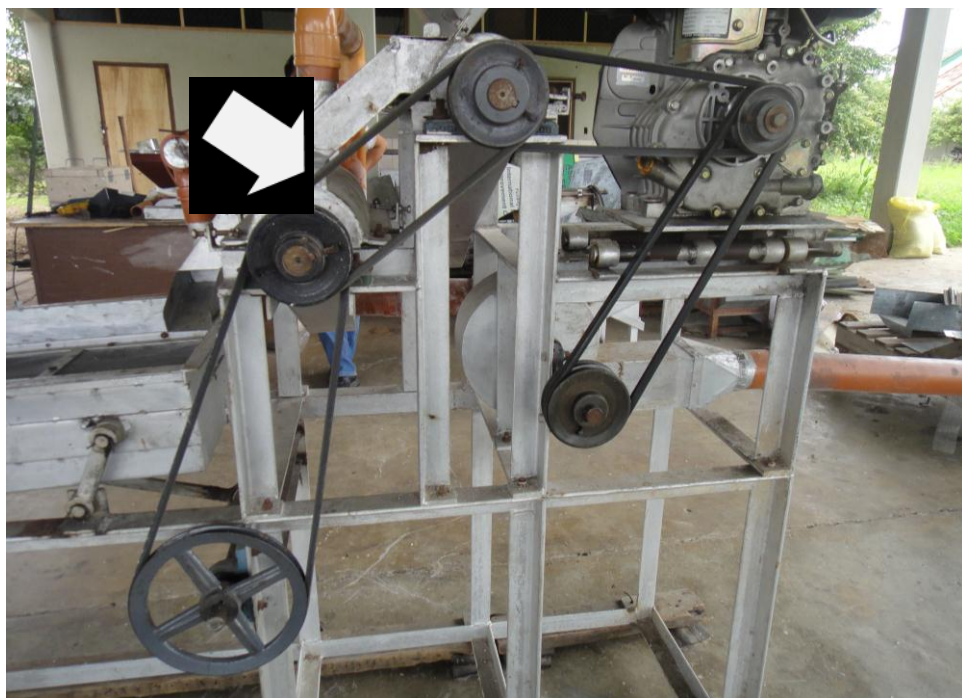


Figure 12. No belt guard

PERFORMANCE TEST

Machine Tested : BARANGAY Corn Mill
Model: CM-250

Date of Test : February 7, 2013

Location of Test : Institute of Plant Breeding
U. P. Los Baños

Table 1. Laboratory analysis of shelled corn samples used in the test

ITEMS	DATA
Variety	IPB Var 6
Source	IPB, Los Baños, Laguna
Kernel Dimensions (mm)	
Length	10.2
Width	7.7
Thickness	4.2
Corn Bulk Density (kg/m ³)	761.0
Moisture Content (%)	12.5
Purity (%)	99.76

Table 3. Machine performance test results

ITEMS	DATA ¹
1. Total weight of input corn kernels (kg)	45.0
2. Duration of input corn kernel feeding (h)	0.24
3. Total weight of corn grits (main product) output (kg)	30.9
4. Duration of corn grits output discharge (h)	0.24
5. Total operating time (h)	0.25
6. Input capacity (kg/h)	187.5
7. Output capacity (kg/h)	128.8
8. Milling capacity (kg/h)	180.0
9. Total weight of product from 1 st screen (main product) output chute	30.9
10. Total weight of product from 2 nd screen (“binlid”) output chute	0.8
11. Total weight of product from bran output chute	9.8
12. Main product recovery (%)	68.7
13. Total weight of by-product (kg)	10.6
14. By-product recovery (%)	23.6
15. Losses (%)	7.7
16. Degerminator Efficiency (%)	5.3

¹ Average of two test trials

Table 3. Machine Performance Test Results (Cont'n.)

ITEMS	DATA ²
17. Analysis of samples from degerminator output chute	
a) Degerminated corn kernels, (%)	5.3
b) Undegerminated whole kernels (%)	0
b) Undegerminated broken kernels (%)	93.2
c) Bran/germs (%)	1.5
18. Analysis of samples from bran outlet	
a) Degerminated corn kernels, (%)	16.9
b) Undegerminated broken kernels (%)	54.1
c) Bran/germs (%)	29.0
19. Analysis of samples from 1 st screen (main product) output chute	
a) Grit #10 (%)	80.4
b) Grit #12 and #14 (%)	16.8
c) Grit #16 and #18 (%)	2.6
d) Grit #20 and #24 (%)	0.1
e) Floured corn, germ and bran (%)	0.1
20. Analysis of samples from 2 nd screen (“binlid”) output chute	
a) Grit #10 (%)	0
b) Grit #12 and #14 (%)	0
c) Grit #16 and #18 (%)	11.9
d) Grit #20 and #24 (%)	46.8
e) Floured corn, germ and bran (%)	41.3
21. Analysis of samples from fan output chute	
a) Grit #10 (%)	26.5
b) Grit #12 and #14 (%)	39.7
c) Grit #16 and #18 (%)	22.5
d) Grit #20 and #24 (%)	4.6
e) Floured corn, germ and bran (%)	6.7

² Average of two test trials

Table 3. Machine Performance Test Results (Cont'n.)

ITEMS	DATA ³
22. Speed of components (rpm) a) Prime mover - Without load - With load b) First Milling Unit (rpm) - Without load - With load c) Second Milling Unit (rpm) - Without load - With load d) Oscillating screen shaft - Without load - With load e) Aspiration fan shaft - Without load - With load	3020 2866 2166 2030 2130 2006 1076 1006 2149 2032
23. Noise Level, [db(A)] a) Feeder - Without load - With Load c) Bagger - Without load - With load	92.8 97.4 91.4 94.2
24. Primemover's fuel consumption (SUPER KAMA KM178F diesel engine), L/h	0.86

³ Average of two test trials

Annex 1. Formula used during calculation and testing

-
1. Input capacity, C_i (kg/h)

$$C_i = \frac{\text{Wt.of corn kernel input, kg}}{\text{Total loading time, h}}$$

2. Output capacity, C_o (kg/h)

$$C_o = \frac{\text{Wt.of main product, kg}}{\text{Output time, h}}$$

3. Milling capacity, C_m (kg/h)

$$C_m = \frac{\text{Wt. of corn kernel input, kg}}{\text{Total operating time, h}}$$

4. Main product recovery, R_m (%)

$$R_m = \frac{\text{Wt.of main product, kg}}{\text{Wt.of input, kg}} \times 100$$

5. By-product recovery, R_b (%)

$$R_b = \frac{\text{Wt.of by - product, kg}}{\text{Wt. of input, kg}} \times 100$$

6. Losses, L (%)

$$L = 100\% - (\text{main product recovery, \%} + \text{by-product recovery, \%})$$

7. Fuel Consumption (L/h)

$$F_c = \frac{\text{Amount fuel consumed, L}}{\text{Total operating time, h}}$$

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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